# Collections Intelligence Platform

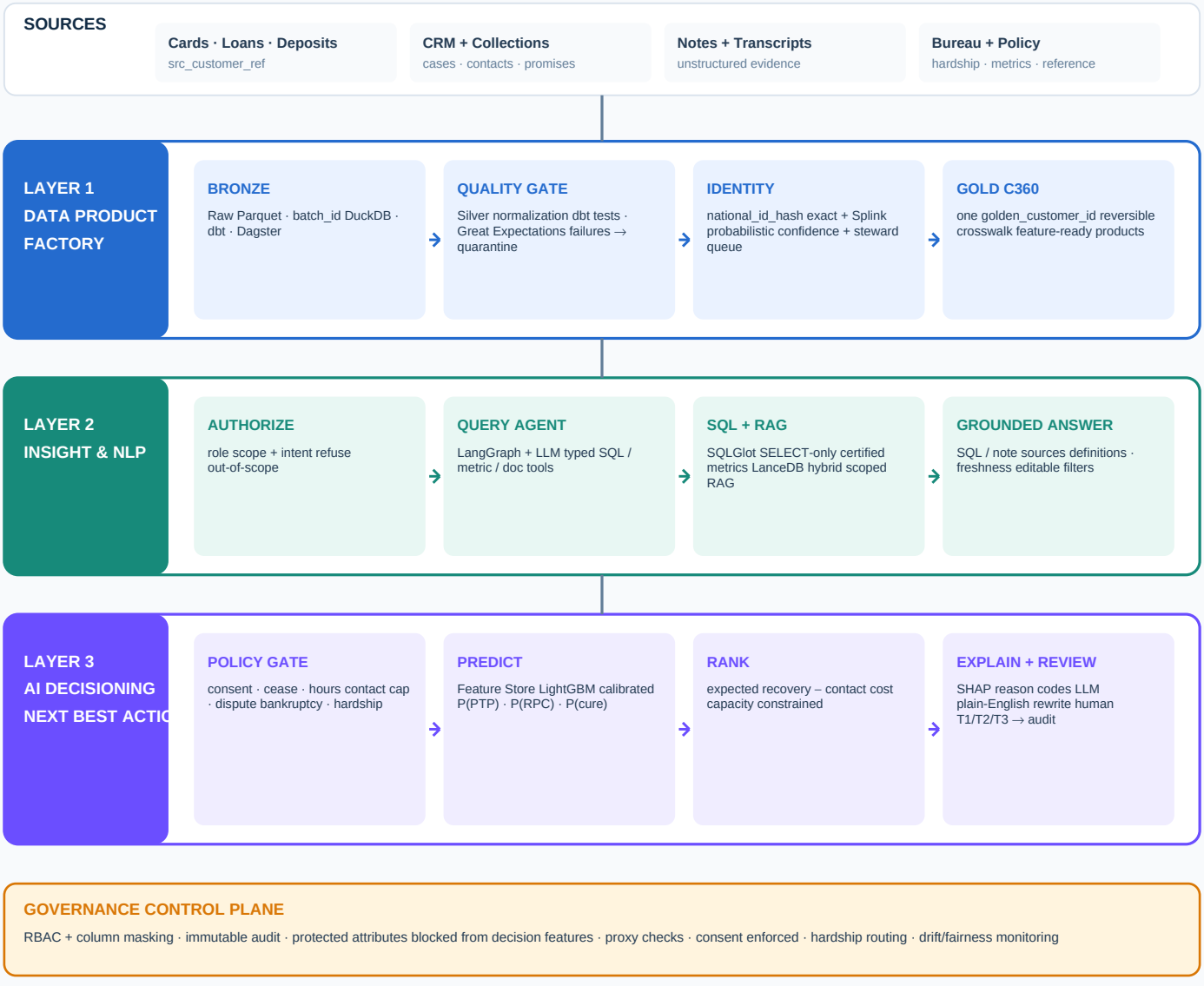
One trusted customer record, plain-English answers that show their SQL, and an explainable Next Best Action that a human can always review.

TEAM: HACKERS MEMBERS: Sathwik · Vishal · Vishnu

31 source tables → 1 golden customer (C360)

Every answer shows SQL or source

Hardship & vulnerability → human




Legend: solid flow = data/decision path | steward/agent/specialist = human review | amber rail = cross-cutting governance

# Summary, Business Value & Layer 1

### PROBLEM

Agents rebuild the customer story across products, CRM, notes and calls; contact becomes slow, inconsistent and sometimes poorly timed.

### USE CASE

Explainable Next Best Action for treatment, channel and timing; hardship and vulnerability are routed to people.

### USERS

Collections agents · team managers · analysts

## BUSINESS VALUE & MEASUREMENT

**PRIMARY: PTP kept rate + 30-day cure rate**

**PROPOSED PILOT CRITERIA — ASSUMPTION: +3pp PTP kept · -10% attempts/cure · no complaint rise**

Baseline: promises_to_pay.ptp_status + collections_cases.cure_flag by product/bucket; current strategy via strategy_code/test_cell.

Measure offline on time-held-out data for discrimination/calibration. Claim uplift only from a controlled NBA-vs-current-strategy pilot on matched eligible cases.

## LAYER 1 — ONE GOLDEN CUSTOMER, WHERE A WRONG MERGE IS HARDER THAN A MISSED MERGE

**Product tables: src_customer_ref** | **CRM / cases / contacts / promises: crm_customer_id** | **Bureau: bank_subject_ref**

~1.02M CRM rows for ~1M customers → duplicates exist even inside CRM.

### 1 BRONZE

Replayable source truth Parquet + batch_id

### 2 SILVER

Normalize DOB, postal, phone → ISO / A1A 1A1 / E.164

### 3 IDENTITY

Exact national_id_hash; Splink name/DOB/phone/email/address

### 4 CROSSWALK

versioned + reversible method · confidence · state · version

### 5 GOLD

C360 + feature tables one golden_customer_id

## CONFIDENCE-AWARE MATCHING

- ≥ 0.95 auto-link
- 0.80–0.95 → data steward review
- < 0.80 stays separate; thresholds tuned on build data

## QUALITY + AI ASSIST

- dbt + Great Expectations before publish; failures quarantined
- LLM suggests mappings and dbt test skeletons
- Schema hash detects drift; LLM explains impact; human approves PR

**Differentiator: confidence-aware identity + reversible crosswalk + provenance. A shared phone/address alone never merges two people.**

# Layer 2 — Insight & NLP

Authorize first, then answer, and always show the working.




## THREE QUERY MODES

- Structured: SQL over governed Gold views
- Unstructured: RAG over agent_notes + call_transcripts
- Hybrid: SQL selects cohort → RAG searches only inside it

## EVIDENCE + SAFETY

- Filter redaction_status, pii_redaction_applied, retention_until
- Retrieved text is untrusted data, never instructions
- Every unstructured claim cites note_id or transcript turn

## ANSWER CARD — SHOW THE WORKING

**Understood: bucket=30–59 DPD · product=unsecured**

**Answer: certified roll-rate metric + concise interpretation**

```
SQL: SELECT … FROM gold.collections_case_metrics …
```

Sources: gold view / note_id / transcript turn · Definition: certified metric · Freshness: as_of_ts

## REFUSE — WITH A SAFE ALTERNATIVE

- Protected-attribute profiling for decisions
- Bulk PII export
- Take customer action directly
- Legal / medical advice
- Outside Collections or beyond role

**benchmark_questions scores should_refuse + refusal_reason.**

## EVALUATION

Execution accuracy | Refusal accuracy | Adversarial/injected notes | Citation completeness | Latency

"Top drivers of roll-rate deterioration" is presented as a descriptive breakdown, explicitly labelled non-causal.

# Layer 3 — AI Decisioning / Next Best Action

Rules decide what is allowed. Models predict what works. A human decides what happens.

1

### DETERMINISTIC POLICY GATE

Consent call/sms/email/push/autodialer · cease request · permitted hours · contact cap TBC · dispute/complaint · bankruptcy/consumer proposal · deceased/fraud · hardship_programs eligibility

2

### POINT-IN-TIME FEATURE STORE

current_dpd · bucket_path · promises_broken · prior_broken_ptp_count · contacts_prior_7d · RPC by channel/time · utilization · payroll delay · risk_score_delta_90d; text features only with measured extraction accuracy

3

### CALIBRATED LIGHTGBM MODELS

P(promise kept) · P(right-party contact by channel/time) · P(cure in 30d). Time-based split; point-in-time snapshots; incumbent model_scores baseline; exclude uses_protected_attributes=true.

4

### CONSTRAINED RANKER

Expected recovery – contact cost, subject to channel_capacity and agent_shifts so the recommendation is executable today.

5

### EXPLAIN

SHAP top-3 reason codes. LLM may rewrite only those codes; checker rejects unsupported reasons. Show alternatives, scores, policy checks and evidence.

6

### HUMAN REVIEW + AUDIT

T1 recommendation surfaced to agent and QA sampled; execution follows approved human-review policy. T2 calls/payment plans/settlements require agent approval. T3 hardship/vulnerability/dispute/complaint → specialist; suppress standard escalation until reviewed.

### KNOWN LIMITATION — HISTORICAL POLICY BIAS

Outcomes exist only for actions that were actually taken. Offline back-tests therefore do not prove uplift; business uplift is claimed only from a controlled pilot.

# AI Usage, Scalability, Governance & 36h Build Plan

## AI USAGE — CLEAR BOUNDARIES

| | | |
|---|---|---|
| **L1 mapping/tests** | LLM suggests | human + deterministic publish |
| **L1 identity** | probabilistic match | no below-threshold auto-merge |
| **L2 NL→SQL** | LLM plans/writes | validator/permissions execute |
| **L2 RAG** | finds/summarises | access rules deterministic |
| **L3 text features** | extracts signals | no final hardship decision |
| **L3 models** | predict outcomes | never override consent/policy |
| **Explanation** | rewrite SHAP codes | never invent reasons |

## SCALABILITY & FAIL-SAFE

**Scale: ~1M customers · 1.84M accounts · 2.5M contacts**

**~1.5M notes · ~20M instalments**

- Nightly: identity/C360, feature backfills, training/open-case scoring, DQ/drift, embeddings
- Near-real-time: payment/PTP/contact events refresh one row + one-case re-score
- Targets <2s score and p95 <10s questions — ASSUMPTIONS
- Fail-safe: stale nightly recommendation + rule baseline if live scoring is down

## GOVERNANCE & RESPONSIBLE AI

- RBAC: agent assigned cases; manager team; analyst aggregate/masked PII; steward; fairness auditor
- Authorize before retrieval; immutable audit of questions, SQL, evidence and decisions
- Protected fields blocked from feature registry; proxies excluded by default / sign-off TBC
- Protected attributes only in restricted fairness-audit environment, never decision models/agents
- Hardship → specialist; monitor PSI, calibration, override, complaint and LLM-feature accuracy

## 36-HOUR BUILD PLAN

| | |
|---|---|
| **24–32** | Integrated C360 + Ask + NBA + review · freeze h28 |
| **32–36** | Benchmark · fairness/quality · demo · repo/docs |

**Must build: C360+identity/DQ · Ask+SQL/refusal · NBA+policy/reasons/human review.**

**Fallback: rules + incumbent score if model misses bar.**

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# collections_customer_360 v0.1

**OWNER: Head of Collections Data Product (named owner TBC) · STEWARD: Collections data steward (TBC) · GRAIN: one row per golden_customer_id**

Purpose: trusted customer context for Collections operations, analytics and approved AI.

| FIELD | TYPE | SENSITIVITY | RULE / PURPOSE |
|---|---|---|---|
| **golden_customer_id** | string PK | Internal | non-null, unique |
| **source_identity_links** | array | Internal | system,id,method,confidence,state |
| **min_match_confidence** | float | Internal | 0..1 |
| **crm_customer_ids** | array<string> | Internal | ≥1 |
| **segment / language / time_zone** | enum | Internal | controlled values |
| **consent_call/sms/email/push/autodialer/letter** | boolean | Internal | non-null; missing = no consent |
| **open_case_ids / bucket / max_dpd / overdue** | mixed | Internal | DPD, overdue ≥0 |
| **promises_kept_90d / promises_broken_90d** | int | Internal | ≥0 |
| **last_contact_ts / last_rpc_ts** | timestamp | Internal | not in future |
| **hardship_flag / hardship_signal** | bool / enum | Sensitive | drives T3 specialist routing |
| **bureau_risk_score / delta_90d** | int | Sensitive | valid score range |
| **dq_status / as_of_ts / lineage_ref** | mixed | Internal | required on every row |

## QUALITY RULES — TESTED AT EVERY PUBLISH

1. golden_customer_id unique + non-null: 100%
2. each source key → at most one golden; <0.80 never auto-merged
3. case/contact/promise resolution ≥99.5%; remainder quarantined
4. range checks 100% or quarantine
5. daily freshness ≤24h; event ≤15m initial SLA TBC; stale flagged
6. consent fields non-null: 100%

## USE, REFRESH & LINEAGE

- Allowed: agent/manager context; governed SQL/doc query; approved features/scoring; audit/governance.
- Prohibited: protected/proxy decision features; unscoped PII; quarantined/stale/unresolved identity; marketing; customer-impacting action without human review capability.
- Refresh: nightly full target 06:00 local TBC + incremental case/contact/payment/PTP target ≤15m TBC.
- Lineage: schema version + source batch + transformation + identity + feature/model version reconstruct every answer/decision.

**Names, contact details and address are served through masked role views. Protected attributes are not in C360; they exist only in a restricted audit table.**